# YouTube-DNN-RecSys — 系统架构 · 召回数据流

Python / Spark data preparation → TFRecords → dnn.py → user & item vectors → prediction

主数据流　　虚线框 = 架构分区

回退 / 反馈　　盒内 = 模块 + 职责

## DATA PROCESSING · 离线样本构建

**Python Pipeline**
data2tfrecords.py
read_tfrecords.py

**Spark Pipeline**
MakeDataOne.scala
MakeDataTwo.scala

**FeatureBuilder**
history + continuous features

## STORAGE · 训练与模型产物

**TFRecords**
train / evaluation

**Checkpoints**
ckpt / eval / modelpath

## MODEL TRAINING · src/python/models/dnn.py

**Input**
visit_items_index × 5
continuous_features × 16

**Item Embedding**
500 × 16 · average pooling

**User Tower**
Dense 64 → 32 → vector 16

**Softmax Output**
500 classes · Top-K

evaluation feedback

## PREDICTION · serving paths

**Python SavedModel**
load_dnn_model.py

**User Vector**
PredictUserVector.scala

**Item Table**
ItemEmbeddingPredictor.scala

## RECOMMENDATION OUTPUT

**Candidate Recall**
user vector × item embedding

**Top-K Items**
next-item recommendation

主链：行为历史 → TFRecords → embedding mean-pool → user vector → item table matching → Top-K。